

Sara Robin Wilson

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Education

Michigan State University Doctor of Philosophy in Computational Mathematics, Science, and Engineering <i>Advised by Prof. Andrew Christlieb and Prof. Brian O'Shea</i>	East Lansing, MI Expected May 2031
University of Pittsburgh Bachelor of Science in Mathematics <i>Cum Laude</i>	Pittsburgh, PA May 2026

Research Interests

plasma physics; numerical analysis of differential equations; scientific machine learning

Honors & Awards

Edna M. Heck Scholarship
Society of Women Engineers Certificate of Merit in Math and Science

Publications

An Analysis of a 2×2 Keyfitz-Kranzer Type Balance System with Varying Generalized Chaplygin Gas <i>Physics of Fluids</i> J. Frew, N. Keyser, E. Kim, G. Paddock, C. Tumbleston, S. Wilson , C. Tsikkou 10.1063/5.0231413 	Sept 2024
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Research Experience

Michigan State University Graduate Student Researcher Research on computational plasma physics.	East Lansing, MI Jun 2026 - Current
Semiconductor Research Corporation REU Research on design modifications for wireless power transfer.	Tuscaloosa, AL Jun 2025 – Jul 2025
West Virginia University REU Research on numerical analysis of 2×2 Keyfitz-Kranzer type balance system with varying Chaplygin gas.	Morgantown, WV Jun 2024 – Jul 2024

Skills

Programming & Scientific Computing: Python; MATLAB; Java
Computational Methods: finite element methods; finite differences; scientific machine learning
Tools & Software: Git; LaTeX; ParaView

Talks & Presentations

Directed Reading Program, Carnegie Mellon University <i>Computational Plasma Physics</i>	Pittsburgh, PA Dec 2025
Summer Research Symposium, University of Alabama <i>Wireless Power Transfer</i>	Tuscaloosa, AL Jul 2025
Pi Mu Epsilon Poster Session, Joint Mathematics Meetings <i>Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System</i>	Seattle, WA Jan 2025
Summer Research Symposium, West Virginia University <i>Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System</i>	Morgantown, WV Jul 2024

Additional Training

Directed Reading Program, Carnegie Mellon University <i>Computational Plasmas Physics & Nodal Discontinuous Galerkin Methods</i>	Pittsburgh, PA Sept 2025 - Apr 2026
Summer School, Princeton University <i>Fluids and Computer Assisted Proofs</i>	Princeton, NJ Aug 2025
Summer School, Texas A&M University <i>Modeling and Simulation of Partial Differential Equations</i>	College Station, TX May 2025

Teaching Experience

University of Pittsburgh Teaching Assistant	Pittsburgh, PA Aug 2024 – Apr 2026
University of Pittsburgh Mathematics Tutor	Pittsburgh, PA Jan 2023 – Apr 2026
West Virginia University REU Mentor	Remote Jun 2025 - Jul 2025